Blessed G/Michael Catholic Secondary School

Semester 1 Agriculture Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which of the following is the most water-efficient modern irrigation system delivering water directly to root zones?

A. Flood irrigation **B.** Furrow irrigation

C. Drip (trickle) irrigation **D.** Sprinkler irrigation

____ **2.** The three primary essential plant macronutrients supplied by standard NPK fertilizers are:

A. Nitrogen, Phosphorus, and Potassium **B.** Nickel, Platinum, and Krypton

C. Nitrogen, Potassium, and Calcium **D.** Sodium, Phosphorus, and Potassium

____ **3.** Integrated Pest Management (IPM) prioritizes which strategy before applying chemical synthetic pesticides?

A. Immediate chemical saturation
B. Biological controls, crop rotation, and cultural pest-resistant varieties
C. Burning all adjacent vegetative fields
D. Flooding crop furrows permanently

____ **4.** Which soil order is prevalent in humid tropical and subtropical regions characterized by deep red horizons rich in iron and aluminum oxides?

A. Oxisols / Ferralsols **B.** Aridisols

C. Gelisols **D.** Histosols

____ **5.** In animal husbandry, what is the term for the feed ration that supplies only the necessary nutrients to keep an animal alive and healthy without gaining or losing weight?

A. Maintenance ration **B.** Production ration

C. Lactation ration **D.** Fattening ration

____ **6.** Which disease affecting Ethiopian teff, wheat, and barley is caused by fungal rust pathogens of the genus Puccinia?

A. Stem and stripe rust **B.** Bacterial blight

C. Mosaic virus **D.** Downy mildew

____ **7.** What is the primary objective of watershed management in agricultural highlands?

A. Conserving soil, harvesting surface rainwater, and rehabilitating degraded catchments
B. Constructing concrete highways
C. Draining all groundwater into oceans
D. Applying synthetic herbicides across hilltops

____ **8.** Which ruminant stomach compartment is called the "true stomach" where gastric juices and hydrochloric acid digest feed chemically?

A. Abomasum **B.** Rumen **C.** Reticulum **D.** Omasum

____ **9.** What is hydroponics in modern commercial agriculture?

A. Cultivating plants in nutrient-rich water solutions without soil
B. Growing crops submerged underwater in lakes
C. Watering crops with salty seawater
D. Planting trees on dry desert dunes

____ **10.** Which grafting method joins a scion twig onto an established rootstock stem?

A. Whip and tongue grafting **B.** Budding only

C. Layering **D.** Stem cutting

____ **11.** What causes soil salinization in poorly drained irrigated arid agricultural lands?

A. Evaporation of shallow mineral-laden groundwater leaving dissolved salts on the soil surface
B. Heavy tropical rainfall leaching minerals
C. Excessive earthworm activity
D. Adding organic compost

____ **12.** Which method of silage preservation maintains fermented forage quality for livestock during dry seasons?

A. Anaerobic fermentation by lactic acid bacteria in sealed silos or pits
B. Drying hay in direct sunlight
C. Freezing in commercial refrigerators
D. Boiling green forage in large vats

____ **13.** Which insect vector transmits African Animal Trypanosomiasis (Nagana) affecting livestock in southwest Ethiopia?

A. Tsetse fly (Glossina species)
B. Mosquito (Anopheles)
C. Tick
D. Housefly

____ **14.** In farm financial management, what does the Benefit-Cost Ratio (BCR) indicate for an investment project?

A. The ratio of discounted total benefits to discounted total costs (viable when BCR > 1)
B. Gross profit before taxes
C. Loan interest rate only
D. Depreciation value of tractors

____ **15.** What is the function of the Broad Bed and Furrow (BBF) technology developed for Ethiopian Vertisols?

A. To drain excess surface rainwater while retaining subsoil moisture for early planting
B. To harvest wind energy
C. To eradicate aphids
D. To spray liquid fertilizer aerially

____ **16.** Which post-harvest loss prevention technology hermetically seals cereal grains against weevils and mould without chemical insecticides?

A. PICS (Purdue Improved Crop Storage) triple-layer bags
B. Woven jute sacks
C. Open wooden cribs
D. Concrete floors

____ **17.** The agricultural process of raising queen bees and honey bee colonies for honey and beeswax is termed:

A. Apiculture **B.** Sericulture **C.** Pisciculture **D.** Aquaculture

____ **18.** Which of the following is an example of a perennial beverage cash crop indigenous to the southwestern forests of Ethiopia?

A. Coffea arabica
B. Camellia sinensis (Tea)
C. Theobroma cacao
D. Sorghum bicolor

____ **19.** In sustainable agriculture, what is the term for planting two or more crop species simultaneously in close proximity on the same field?

A. Intercropping (polyculture)
B. Monocropping
C. Shifting cultivation
D. Continuous bare fallowing

____ **20.** What is the role of mycorrhizal fungi in plant-soil symbioses?

A. They extend fungal hyphae through soil to vastly increase plant root phosphorus and water uptake
B. They kill crop roots
C. They produce synthetic nitrogen gas
D. They block sunlight from weed seeds